

Crypt Arithmetic Problem Using Python

Aim

To write a Python program to solve the **Crypt Arithmetic Problem** by assigning unique digits to letters such that the given arithmetic equation is satisfied.

Algorithm

1. Start the program.
2. Define the words involved in the crypt arithmetic puzzle.
3. Identify all unique letters in the puzzle.
4. Generate all possible digit assignments for the letters.
5. Ensure that no two letters are assigned the same digit.
6. Check that the leading letter of each word is not assigned zero.
7. Verify whether the arithmetic equation is satisfied.
8. If a valid assignment is found, display the solution.
9. Stop the program.

Python Code

```
from itertools import permutations

letters = "SENDMORY"
digits = range(10)

for p in permutations(digits,
len(letters)):
    d = dict(zip(letters, p))

    if d['S'] == 0 or d['M'] == 0:
        continue

    SEND = 1000*d['S'] + 100*d['E'] +
10*d['N'] + d['D']
    MORE = 1000*d['M'] + 100*d['O'] +
10*d['R'] + d['E']
    MONEY = 10000*d['M'] + 1000*d['O'] +
100*d['N'] + 10*d['E'] + d['Y']

    if SEND + MORE == MONEY:
        print("SEND =", SEND)
        print("MORE =", MORE)
        print("MONEY =", MONEY)
        break
```

EXP 4(Cript-Arithmetic Problem).py > ...

```
1  from itertools import permutations
2  letters = "SENDMORY"
3  digits = range(10)
4  for p in permutations(digits, len(letters)):
5      d = dict(zip(letters, p))
6      if d['S'] == 0 or d['M'] == 0:
7          continue
8      SEND = 1000*d['S'] + 100*d['E'] + 10*d['N'] + d['D']
9      MORE = 1000*d['M'] + 100*d['O'] + 10*d['R'] + d['E']
10     MONEY = 10000*d['M'] + 1000*d['O'] + 100*d['N'] + 10*d['E'] + d['Y']
11     if SEND + MORE == MONEY:
12         print("SEND =", SEND)
13         print("MORE =", MORE)
14         print("MONEY =", MONEY)
15     break
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS E:\OPEN CV\AI Lab> & C:\Users\user\AppData\Local\Programs\Python\Python310\python.exe
SEND = 9567
MORE = 1085
MONEY = 10652
PS E:\OPEN CV\AI Lab> 
```